Dr. HAJRA MOMIN

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OBJECTIVE

Professional transitioning from academia to an industry role as an Artificial Intelligence Developer. Seeking to gain hands-on experience, contribute to real-world AI projects, and collaborate with industry professionals. Committed to leveraging my enthusiasm and eagerness to learn in order to make meaningful contributions and drive impactful results.

EDUCATION

PhD in Computer Science
University of KwaZulu-Natal, South Africa (2010–2017)

Master of Computer Applications (MCA)
Sinhgad Institute of Business Administration and Research, University of Pune (2005–2008)

- First Class with Distinction (71.86%)

Bachelor of Science (Electronics)
Poona College of Arts, Science and Commerce, University of Pune (2002–2005)

- First Class with Distinction (86.63%)

SKILLS SUMMARY

Languages:	Python, Java
Databases/OS:	MySQL, MS-Access, Windows, Linux
Python Libraries:	Pandas, Numpy, Matplotlib, Sci-kit learn, seaborn
Frameworks:	Jupyte, Spyder, Anaconda, Google-Colab
Web:	HTML, XML, JSP, ASP.NET

PROFESSIONAL EXPERIENCE

PGT (Computer Science)
The Kalyani School, Pune (Apr 2022 – May 2023)

- Taught Python, MySQL, AI, and Robotics to senior secondary students.
- Developed project-based learning modules aligned with modern CS education.

TGT (ICT)
Global Indian International School, Pune (Nov 2021 – Apr 2022)

- Delivered ICT curriculum focusing on foundational programming and digital skills.

Coding Instructor (Python, JavaScript)
Purple Tutor (Feb 2021 – Nov 2021)

- Conducted coding sessions for students worldwide; customized learning paths.

PGT (Informatics Practices)
Lexicon International School, Pune (Oct 2018 – Feb 2021)

- Taught database management, data analysis, and Python programming.
- Guided student participation in tech exhibitions and national competitions.

Lecturer (Part-Time)
University of KwaZulu-Natal, South Africa (2010–2015)

- Delivered undergraduate modules in Computer Programming and related subjects.
- Mentored student research projects in machine learning and image processing.
- Actively contributed to departmental seminars and academic initiatives.

Technical Associate
Tech Mahindra (2008 – 2009)

- Worked on invoice discrepancy analysis using SQL and UNIX tools.
- Tested system procedures in HR management modules.

Intern – Software Developer
Octaware Technologies Ltd. (2008)

- Developed an online bookstore shopping cart application using JSP.

AI & MACHINE LEARNING SKILLS

- Machine Learning Algorithms (Supervised, Unsupervised, Reinforcement)
- Deep Learning (Neural Networks, CNNs, RNNs)
- Computer Vision (Image Processing, Facial Recognition)
- Natural Language Processing (Text Classification, Tokenization, NLTK)
- Model Training & Evaluation (Cross-validation, Confusion Matrix)
- Tools & Frameworks (TensorFlow, Keras, Scikit-learn, PyTorch)
- Python for Data Science (NumPy, Pandas, Matplotlib, Seaborn)

CERTIFICATIONS

- *Completed Master's in Artificial Intelligence (Simplilearn + IBM)* — July 2025
- NPTEL: *Introduction to Machine Learning*, IIT Kharagpur – 2017
- OOAD & UML Techniques (Zensoft Certified) – 2007

- Diploma in Computer Applications & Multilingual DTP – 2004

PUBLICATIONS

- **Momin, H.** & Tapamo, J. R. (2011). *Automatic Detection of Face and Facial Landmarks for Face Recognition*.
  International Conference on Signal Processing, Image Processing and Pattern Recognition (SIP 2011), South Korea, pp. 244–253.
- **Momin, H.** & Tapamo, J. R. (2016). *Gabor Filter Based Texture Analysis of Facial Components for Ethnicity Classification*.
  International Journal of Applied Mathematics and Information Sciences, Vol. 10, No. 6, pp. 2255–2265.

PhD RESEARCH SUMMARY

Thesis Title: *Studies in Machine Learning Techniques for Facial-Based Ethnicity Classification*

- Investigated feature-based classification of ethnicity from grayscale facial images.
- Explored color, texture (Gabor filters), and geometric features for classification.
- Applied various machine learning algorithms to compare accuracy across models.
- Ethnic categories studied: Asian, Indian, White, and Black.

ACADEMIC PROJECTS

Web-Based Tour Management System
ASP.NET, MS Access
Developed a web interface for booking and managing travel services, including live tour and flight status updates.

Shopping Cart for Online Books
JSP
Designed a dynamic book sales platform with search, cart, admin, and reporting modules.

Light Intensity Controller Using Microcontroller
Assembly Language
Implemented an embedded system to control lighting intensity, applicable in agriculture and healthcare domains.

PERSONAL DETAILS

Address:	Flat No 103, Span Residency, Magarpatta, Hadapsar, Pune, India
Date of Birth:	15 August 1985
Nationality:	Indian
Marital Status:	Married
Gender:	Female